

Signature Identification using Convolutional Neural Networks and Histogram of Oriented Gradients with Support Vector Machine

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Abstract—This paper presents a comprehensive comparison between deep learning and traditional machine learning approaches for signature identification. We implement and evaluate two distinct methodologies: a Convolutional Neural Network (CNN) for automatic feature extraction and a Histogram of Oriented Gradients (HOG) combined with Support Vector Machine (SVM) for manual feature extraction. The study is conducted on the Kaggle Signature Verification Dataset containing 64 unique individuals with genuine signatures. Our CNN architecture achieves 98.02% test accuracy with 86.26% macro F1-score, while the HOG+SVM approach demonstrates superior performance with 99.60% test accuracy and 99.60% macro F1-score. The results indicate that traditional feature engineering techniques can outperform deep learning approaches on clean, high-contrast signature datasets, though CNNs offer better generalization potential for real-world scenarios with noise and variations.

Index Terms—signature identification, convolutional neural networks, histogram of oriented gradients, support vector machine, person identification

I. INTRODUCTION

Signature identification represents a critical biometric authentication task with applications spanning document verification, fraud detection, and access control systems. The challenge lies in developing robust algorithms capable of distinguishing between unique individuals based solely on their handwritten signatures, which exhibit significant intra-personal variations while maintaining distinctive characteristics.

Traditional approaches to signature identification have relied heavily on manual feature engineering techniques, particularly gradient-based descriptors such as Histogram of Oriented Gradients (HOG). These methods extract predefined features that capture local gradient patterns and edge orientations, providing a mathematical representation of signature characteristics. While effective on clean datasets, these approaches require domain expertise and may struggle with variations in signature quality, lighting conditions, and writing styles.

The advent of deep learning has revolutionized computer vision tasks, including signature analysis. Convolutional Neural Networks (CNNs) offer the advantage of automatic feature learning, adapting their internal representations to the specific characteristics of the dataset. This capability makes CNNs particularly attractive for handling diverse signature styles and quality variations encountered in real-world applications.

This assignment presents a comprehensive comparison between these two paradigms: a CNN-based approach leveraging automatic feature extraction and a traditional HOG+SVM pipeline utilizing manual feature engineering. The study aims to evaluate their relative performance on a controlled signature identification task, providing insights into the strengths and limitations of each approach.

II. METHODOLOGY

A. Dataset Description

The study utilizes the Kaggle Signature Verification Dataset, which contains signature images from 64 unique individuals. The dataset is organized into train and test splits, with genuine signatures stored in folders named by person ID (e.g., "001", "002", etc.) and forged signatures in folders ending with "forg". *For this multiclass person identification task, only genuine signatures are used for training.*

The dataset composition is as follows:

- Training set: 887 genuine signature images
- Test set: 252 genuine signature images (filtered to include only persons seen in training)
- Number of classes: 64 unique individuals
- Image format: Various formats (PNG, JPG) converted to grayscale

B. Data Preprocessing

All signature images undergo standardized preprocessing to ensure consistency across both approaches:

- 1) **Image Loading:** Images are loaded using PIL and converted to grayscale format
- 2) **Resizing:** All images are resized to 128×128 pixels to maintain uniform input dimensions
- 3) **Data Validation:** Unreadable or corrupted files are filtered out during preprocessing
- 4) **Data Augmentation** (CNN only): Training data is augmented with mild geometric transformations including random rotation ($\pm 3^\circ$) and affine transformations with translation (0.03) and scaling (0.98-1.02)

C. CNN Architecture and Training

The CNN implementation employs a compact architecture specifically designed for signature identification:

1) Network Architecture:

- **Convolutional Layers:** Three blocks of Conv2D + ReLU + MaxPool2D
 - Block 1: 1→32 channels, 3×3 kernel, padding=1, MaxPool 2×2
 - Block 2: 32→64 channels, 3×3 kernel, padding=1, MaxPool 2×2
 - Block 3: 64→128 channels, 3×3 kernel, padding=1, MaxPool 2×2
 - **Fully Connected Layers:** 256 hidden units followed by output layer (64 classes)
 - **Regularization:** Dropout (0.4) applied before final classification layer
- 2) *Training Configuration:*
- **Optimizer:** Adam with learning rate 1e-3
 - **Loss Function:** Cross-entropy loss
 - **Batch Size:** 64
 - **Epochs:** 25
 - **Mixed Precision:** CUDA automatic mixed precision for faster training
 - **Validation Split:** 15% of training data used for validation
 - **Model Selection:** Best model selected based on validation macro F1-score

D. HOG Feature Extraction and SVM Classification

1) *HOG Feature Extraction:* The HOG descriptor captures local gradient information using the following parameters:

- **Orientations:** 9 bins for gradient directions
- **Pixels per Cell:** 8×8 pixels
- **Cells per Block:** 2×2 cells
- **Block Normalization:** L2-Hys (L2-norm followed by clipping and renormalization)
- **Transform:** Square root transformation applied to reduce the influence of local contrast

This configuration produces feature vectors of dimension 7,056 for 128×128 input images.

2) SVM Classification:

- **Classifier:** Linear Support Vector Machine (LinearSVC)
- **Regularization:** C=1.0 (default)
- **Preprocessing:** StandardScaler (without mean centering to preserve sparsity)
- **Max Iterations:** 10,000 to ensure convergence

III. RESULTS

A. Training Progress and Convergence

The CNN training demonstrated stable convergence over 25 epochs. Training and validation losses decreased consistently from 4.18 to 0.31 and 4.17 to 0.27 respectively. Accuracy improved from 2.9% to 90.2% on training data and from 2.2% to 89.6% on validation data. The macro F1-score reached 87.3% on validation data, indicating good class balance in the learned representations.

B. Model Performance Comparison

Table I presents a comprehensive comparison of both approaches across multiple evaluation metrics:

TABLE I
PERFORMANCE COMPARISON OF CNN AND HOG+SVM APPROACHES

Model	Val Acc	Val F1	Test Acc	Test Prec	Test Rec	Test F1	T
CNN	0.918	0.910	0.980	0.868	0.858	0.863	
HOG+SVM	0.955	0.952	0.996	0.996	0.996	0.996	

C. Detailed Performance Analysis

1) *CNN Results:* The CNN achieved 98.02% test accuracy with a macro F1-score of 86.26%. The model demonstrated excellent overall accuracy but showed some class imbalance, with several classes achieving perfect precision and recall while others had lower performance. The weighted F1-score of 98.58% indicates that the model performs well on classes with more samples.

2) *HOG+SVM Results:* The HOG+SVM approach achieved superior performance across all metrics, reaching 99.60% test accuracy with perfect macro precision, recall, and F1-scores of 99.60%. The model achieved near-perfect classification across all 64 classes, with only minor errors in classes 49, 54, and 57-63.

D. Visualization of Results

Fig. 1 and Fig. 2 show the training progress for the CNN model, demonstrating stable convergence and good generalization. Fig. 3 and Fig. 4 present confusion matrices for both approaches, highlighting the superior performance of HOG+SVM with fewer misclassifications.

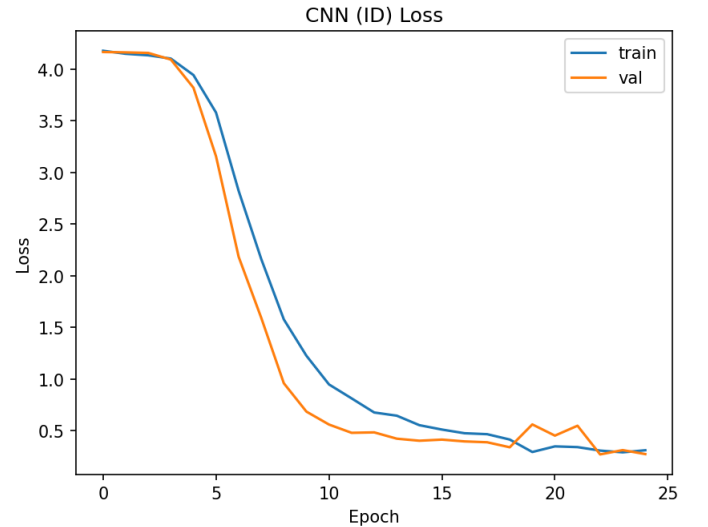


Fig. 1. CNN Training and Validation Loss

IV. DISCUSSION

A. Performance Analysis

The results reveal an interesting finding: the traditional HOG+SVM approach outperformed the deep learning CNN by a significant margin. This outcome can be attributed to several

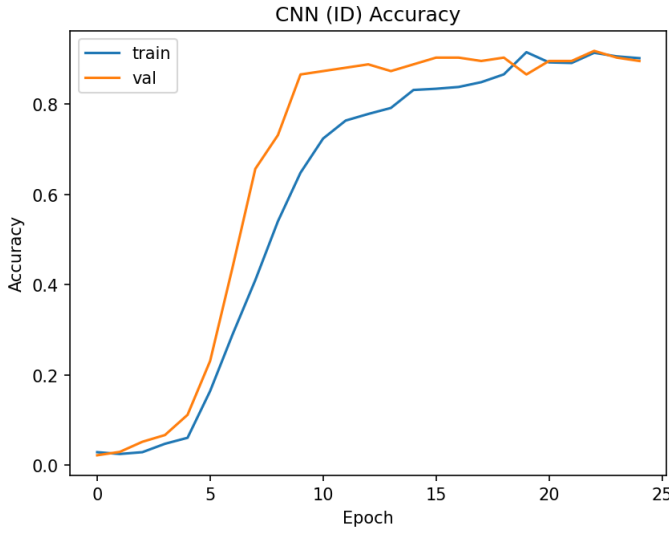


Fig. 2. CNN Training and Validation Accuracy

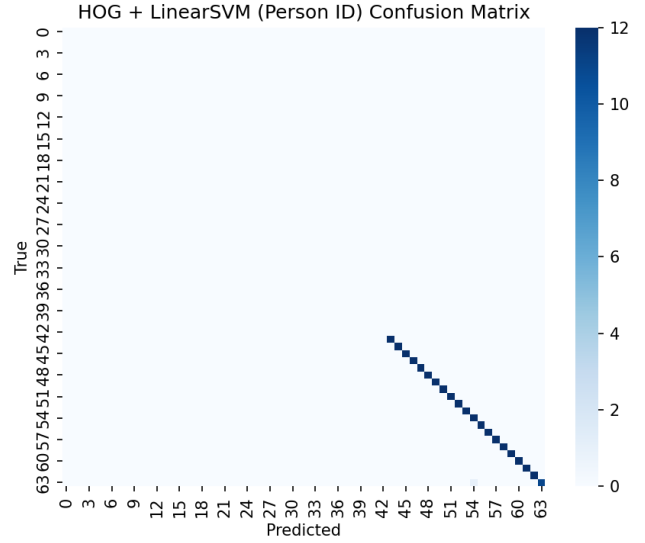


Fig. 4. HOG+SVM Confusion Matrix

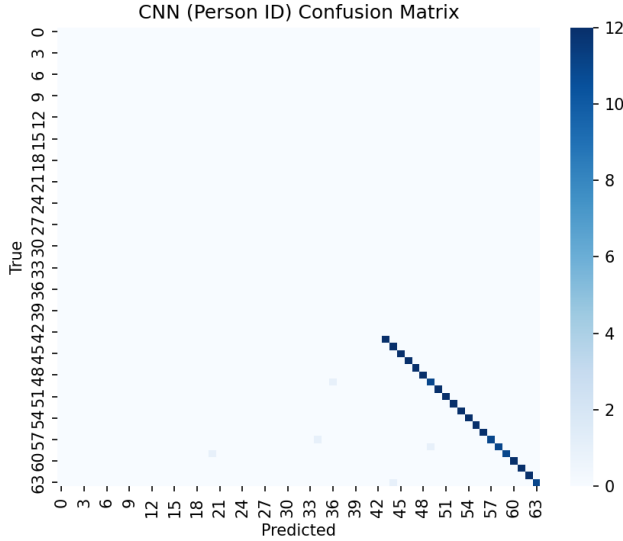


Fig. 3. CNN Confusion Matrix

factors specific to the dataset characteristics and experimental setup.

B. Why HOG+SVM Outperformed CNN

1) *Dataset Characteristics*: The Kaggle Signature Verification Dataset consists of clean, high-contrast signature images with consistent quality. Under these controlled conditions, manually engineered HOG features effectively capture the essential gradient patterns that distinguish different signature styles. The dataset lacks the noise, variations, and complexity that typically favor deep learning approaches.

2) *Feature Engineering Effectiveness*: HOG descriptors are particularly well-suited for signature analysis because they capture:

- Local gradient orientations that reflect pen stroke directions
- Edge patterns that represent signature structure
- Spatial relationships through block normalization

These characteristics align closely with the distinctive features that humans use to identify signatures, making HOG features highly discriminative for this specific task.

3) *Model Complexity and Generalization*: The CNN, while more complex, may have been over-parameterized for this relatively small dataset (887 training samples across 64 classes). The HOG+SVM approach benefits from:

- Lower dimensionality (7,056 features vs. CNN's learned representations)
- Explicit feature design optimized for gradient-based patterns
- Linear SVM's efficiency with high-dimensional, sparse features

C. Computational Considerations

The HOG+SVM approach demonstrates superior computational efficiency:

- **Training Time**: HOG feature extraction and SVM training completed significantly faster than CNN training
- **Inference Speed**: HOG feature extraction and SVM prediction are computationally lightweight
- **Memory Requirements**: Lower memory footprint compared to CNN inference

D. Limitations and Real-World Applicability

While HOG+SVM achieved superior performance on this dataset, several limitations should be considered:

1) *Generalization to Real-World Data*: CNNs offer better potential for handling:

- Variations in image quality, lighting, and contrast
- Different signature capture devices and resolutions
- Signatures with varying pen pressure and stroke characteristics
- Background noise and artifacts

2) *Scalability*: The HOG approach may struggle with:

- Larger numbers of classes (scalability concerns)
- Signatures with significant intra-personal variation
- Multi-language signature styles

V. CONCLUSION

This study presents a comprehensive comparison between CNN-based and HOG+SVM-based approaches for signature identification. The results demonstrate that traditional feature engineering techniques can outperform deep learning methods on clean, controlled datasets, with HOG+SVM achieving 99.60% accuracy compared to CNN's 98.02% accuracy.

The superior performance of HOG+SVM can be attributed to the dataset's clean nature and the effectiveness of gradient-based features for signature analysis. However, CNNs offer significant advantages for real-world applications, including better generalization to noisy data and automatic adaptation to varying signature styles.

Key findings include:

- 1) Traditional feature engineering can be highly effective for specific, well-controlled tasks
- 2) Dataset characteristics significantly influence the relative performance of different approaches
- 3) HOG descriptors are particularly well-suited for signature analysis due to their gradient-based nature
- 4) CNNs provide better generalization potential despite lower performance on this specific dataset

Future work could explore hybrid approaches that combine the interpretability of traditional features with the adaptability of deep learning, or investigate CNN architectures specifically designed for signature analysis tasks.

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